

Computer

Practical

File

2025-'26

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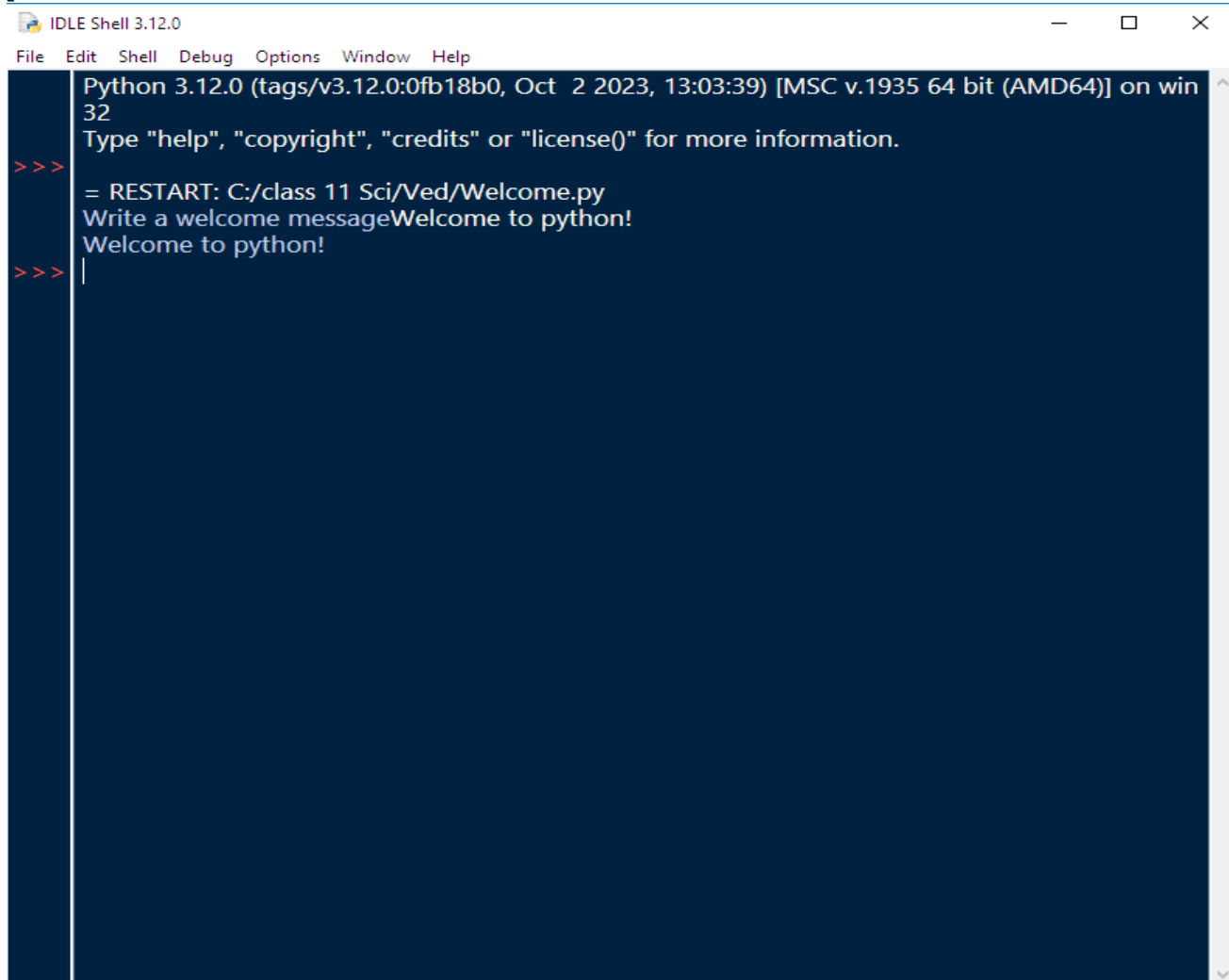
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Program #1- Input a welcome message and display it.

Code:-

```
# Input a welcome message and display it.
a=input("Write a welcome message")
print(a)
```



```
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win
32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/class 11 Sci/Ved/Welcome.py
Write a welcome messageWelcome to python!
Welcome to python!
>>> |
```

Program #2-

PERIMETER AND AREA OF A CIRCLE

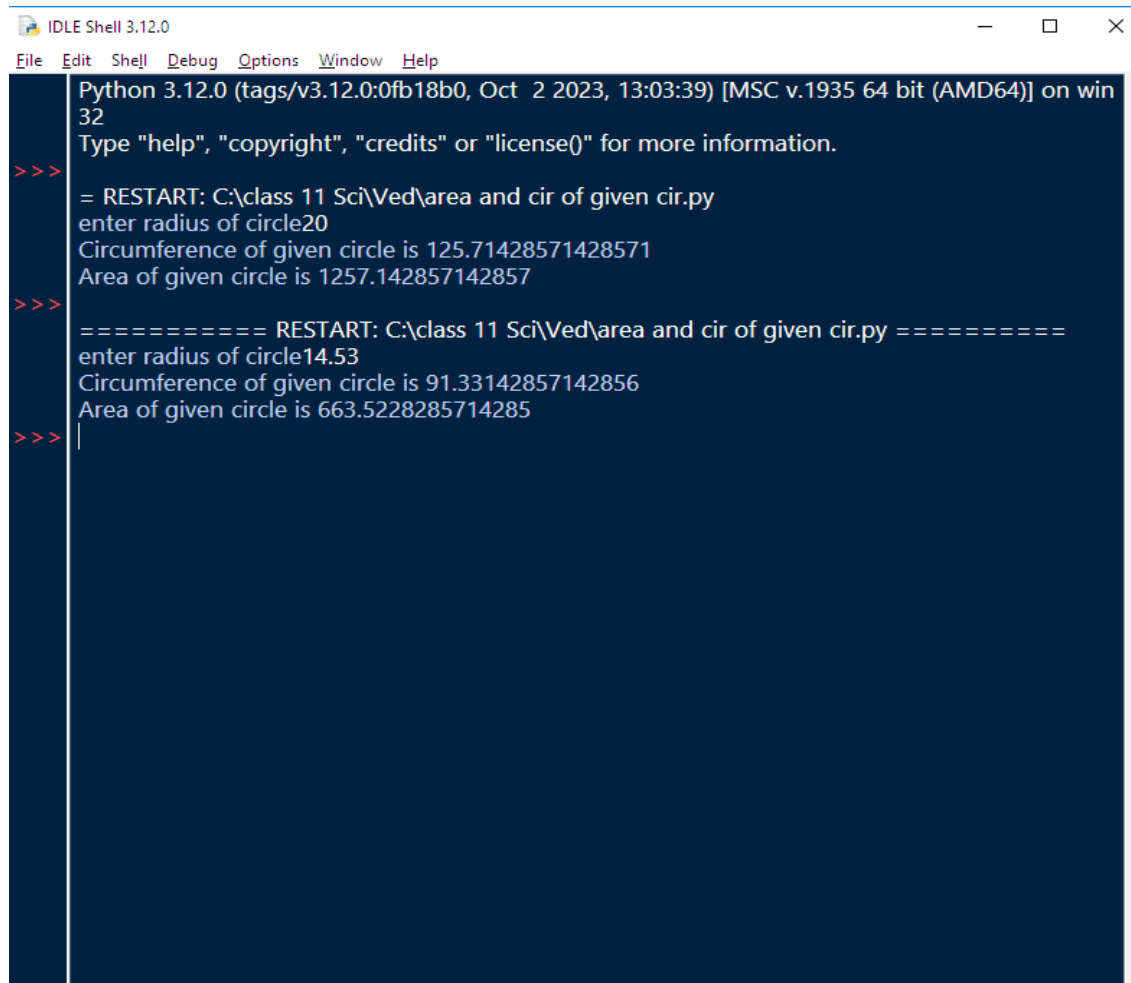
Code:-

```
# PERIMETER AND AREA OF A CIRCLE
r=eval(input('enter radius of circle'))
cir=2*22/7*r
```

```

print('Perimeter of given circle is', cir)
ar=22/7*r**2
print('Area of given circle is', ar)

```



Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

```

>>> = RESTART: C:\class 11 Sci\Ved\area and cir of given cir.py
enter radius of circle20
Circumference of given circle is 125.71428571428571
Area of given circle is 1257.142857142857
>>> ===== RESTART: C:\class 11 Sci\Ved\area and cir of given cir.py =====
enter radius of circle14.53
Circumference of given circle is 91.33142857142856
Area of given circle is 663.5228285714285
>>> |

```

Program #3-

Input three numbers and display the largest / smallest number.

Code:-

```

a=int(input("Enter no. 1"))
b=int(input("Enter no. 2"))
c=int(input("Enter no. 3"))
if a<b and a<c:
    print(a, "is smallest")
elif b<a and b<c:

```

```

        print(b, "is smallest")
elif c<a and c<b:
    print(c, "is smallest")
elif a<=b and a<c:
    print("first and second are smallest")
elif b<a and b<=c:
    print(" second and third are smallest")
elif a<=c and a<b:
    print(" first and third are smallest")
elif a==b and a==c:
    print("all no's are equal")

if a>b and a>c:
    print(a, "is biggest")
elif b>a and b>c:
    print(b, "is biggest")
elif c>a and c>b:
    print(c, "is biggest")
elif a>=b and a>c:
    print("first and second are biggest")
elif b>a and b>=c:
    print(" second and third are biggest")
elif a>=c and a>b:
    print(" first and third are biggest")
elif a==b and a==c:
    print("all no's are equal")

```

```
IDLE Shell 3.12.0
File Edit Shell Debug Options Window Help

= RESTART: C:/class 11 Sci/Ved/equality check 3 numbers.py
Enter no. 11
Enter no. 21
Enter no. 32
first and second are smallest
2 is biggest
>>>

===== RESTART: C:/class 11 Sci/Ved/equality check 3 numbers.py =====
Enter no. 12
Enter no. 21
Enter no. 31
second and third are smallest
2 is biggest
>>>

===== RESTART: C:/class 11 Sci/Ved/equality check 3 numbers.py =====
Enter no. 11
Enter no. 22
Enter no. 33
1 is smallest
3 is biggest
>>>

===== RESTART: C:/class 11 Sci/Ved/equality check 3 numbers.py =====
Enter no. 13
Enter no. 22
Enter no. 33
2 is smallest
first and third are biggest
>>>

===== RESTART: C:/class 11 Sci/Ved/equality check 3 numbers.py =====
Enter no. 13
Enter no. 25
Enter no. 33
first and third are smallest
5 is biggest
>>>

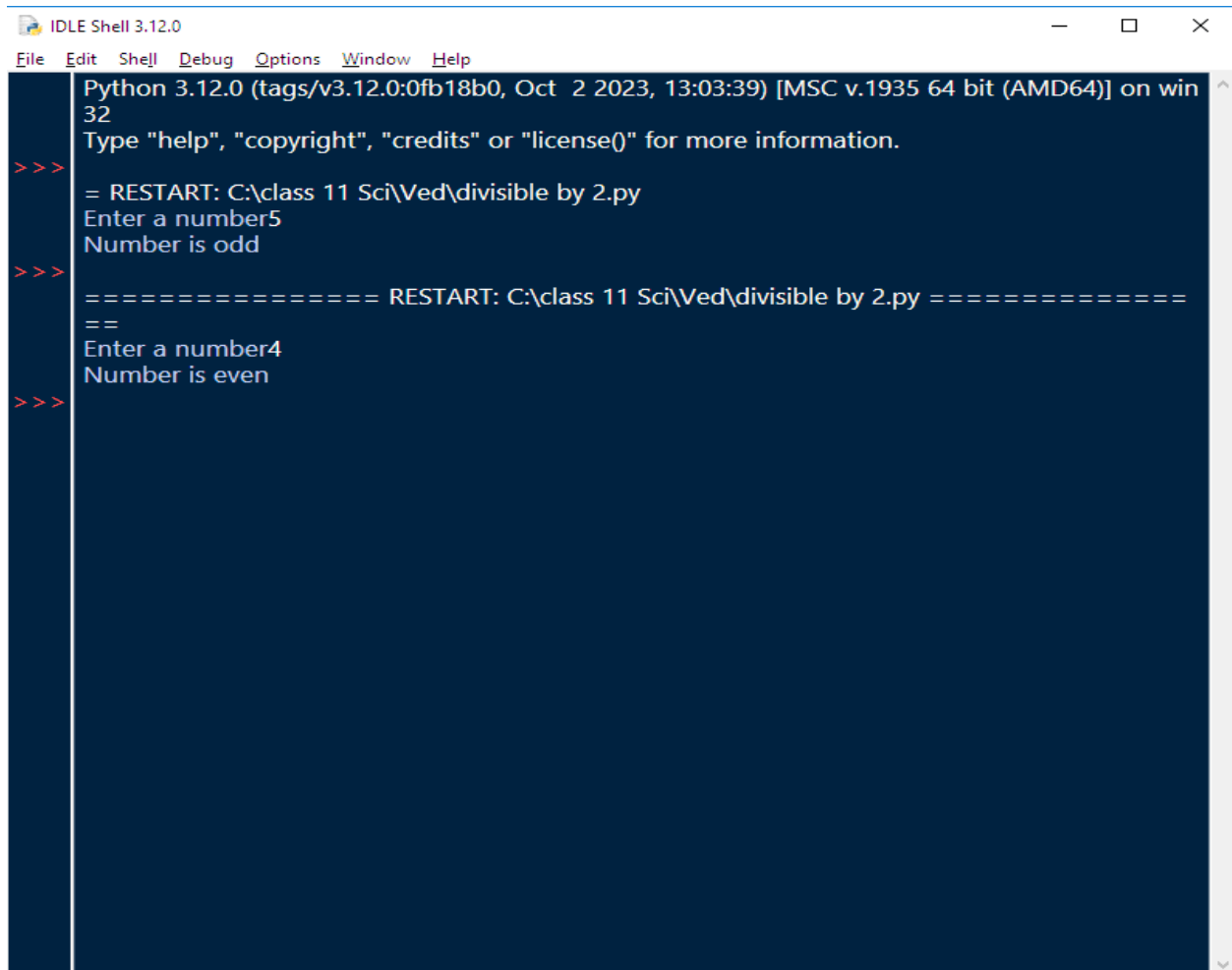
Ln: 38 Col: 0
```

Program #4-

Write a program TO CHECK ENTERED NUMBER IS ODD /EVEN.

Code:-

```
a=int(input("Enter a number"))
if a%2==0:
    print("Number is even")
else:
    print("Number is odd")
```

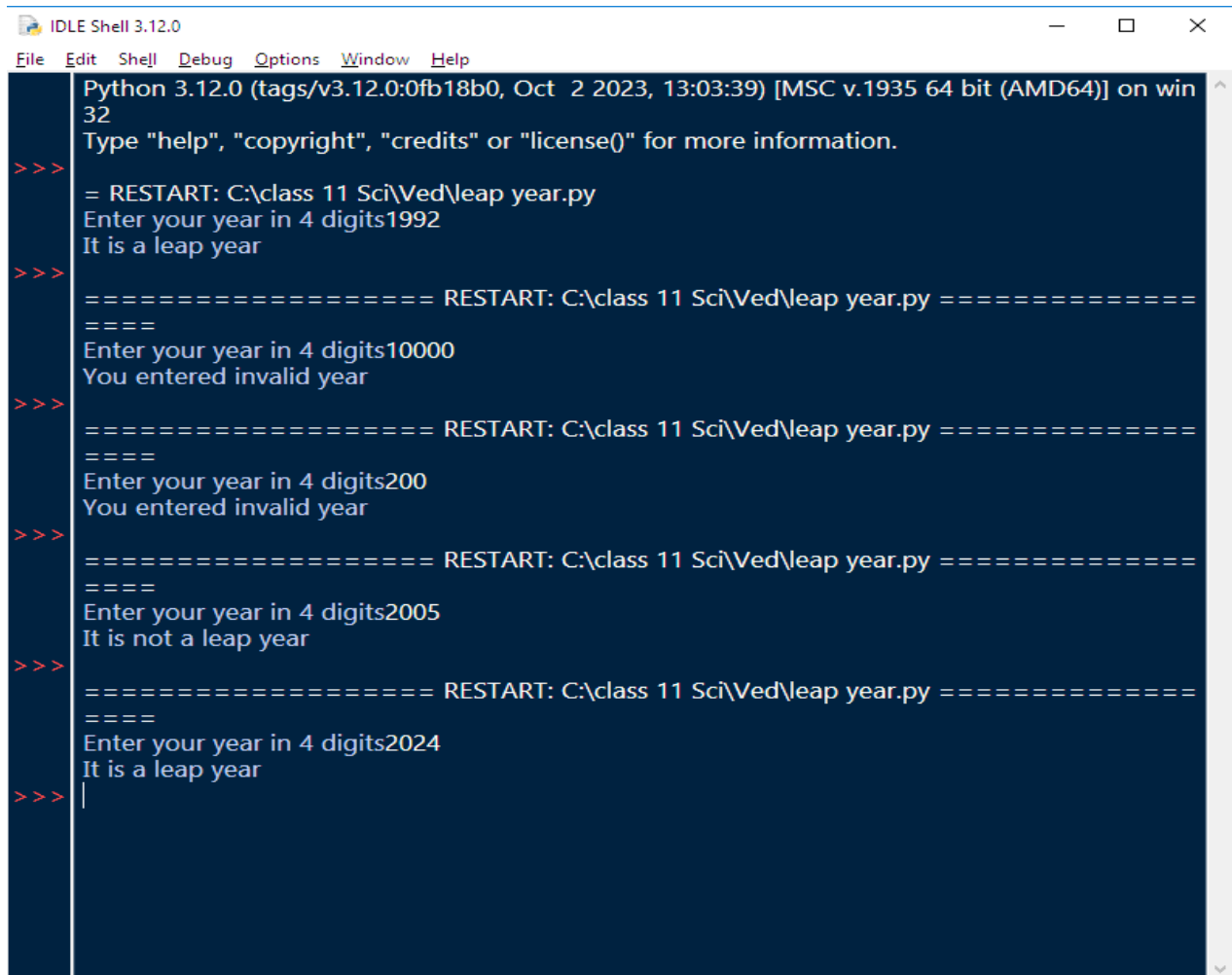
A screenshot of the IDLE Shell 3.12.0 window. The title bar reads 'IDLE Shell 3.12.0'. The menu bar includes 'File', 'Edit', 'Shell', 'Debug', 'Options', 'Window', and 'Help'. The shell area has a dark blue background with white text. It shows the Python 3.12.0 version and architecture (MSC v.1935 64 bit (AMD64)) on Windows 32. It prompts the user to type 'help', 'copyright', 'credits', or 'license()'. There are two execution sessions shown, each preceded by a red '>>>' prompt. The first session shows a restart of a script 'divisible by 2.py', followed by the input '5' and the output 'Number is odd'. The second session shows another restart of the same script, followed by the input '4' and the output 'Number is even'.

```
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win
32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\class 11 Sci\Ved\divisible by 2.py
Enter a number5
Number is odd
>>>
===== RESTART: C:\class 11 Sci\Ved\divisible by 2.py =====
==
Enter a number4
Number is even
>>>
```

Program #5- Write a program to check for LEAP YEAR

Code:-

```
a=int(input("Enter your year in 4 digits"))
if a>999 and a<10000:
    if a%4==0:
        print("It is a leap year")
    else:
        print("It is not a leap year")
else:
    print("You entered invalid year")
```

```
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\class 11 Sci\Ved\leap year.py
Enter your year in 4 digits1992
It is a leap year
>>>
===== RESTART: C:\class 11 Sci\Ved\leap year.py =====
=====
Enter your year in 4 digits10000
You entered invalid year
>>>
===== RESTART: C:\class 11 Sci\Ved\leap year.py =====
=====
Enter your year in 4 digits200
You entered invalid year
>>>
===== RESTART: C:\class 11 Sci\Ved\leap year.py =====
=====
Enter your year in 4 digits2005
It is not a leap year
>>>
===== RESTART: C:\class 11 Sci\Ved\leap year.py =====
=====
Enter your year in 4 digits2024
It is a leap year
>>>
|
```

Program #6-

Write a program to check entered character is vowels, consonants, uppercase, lowercase, special characters.

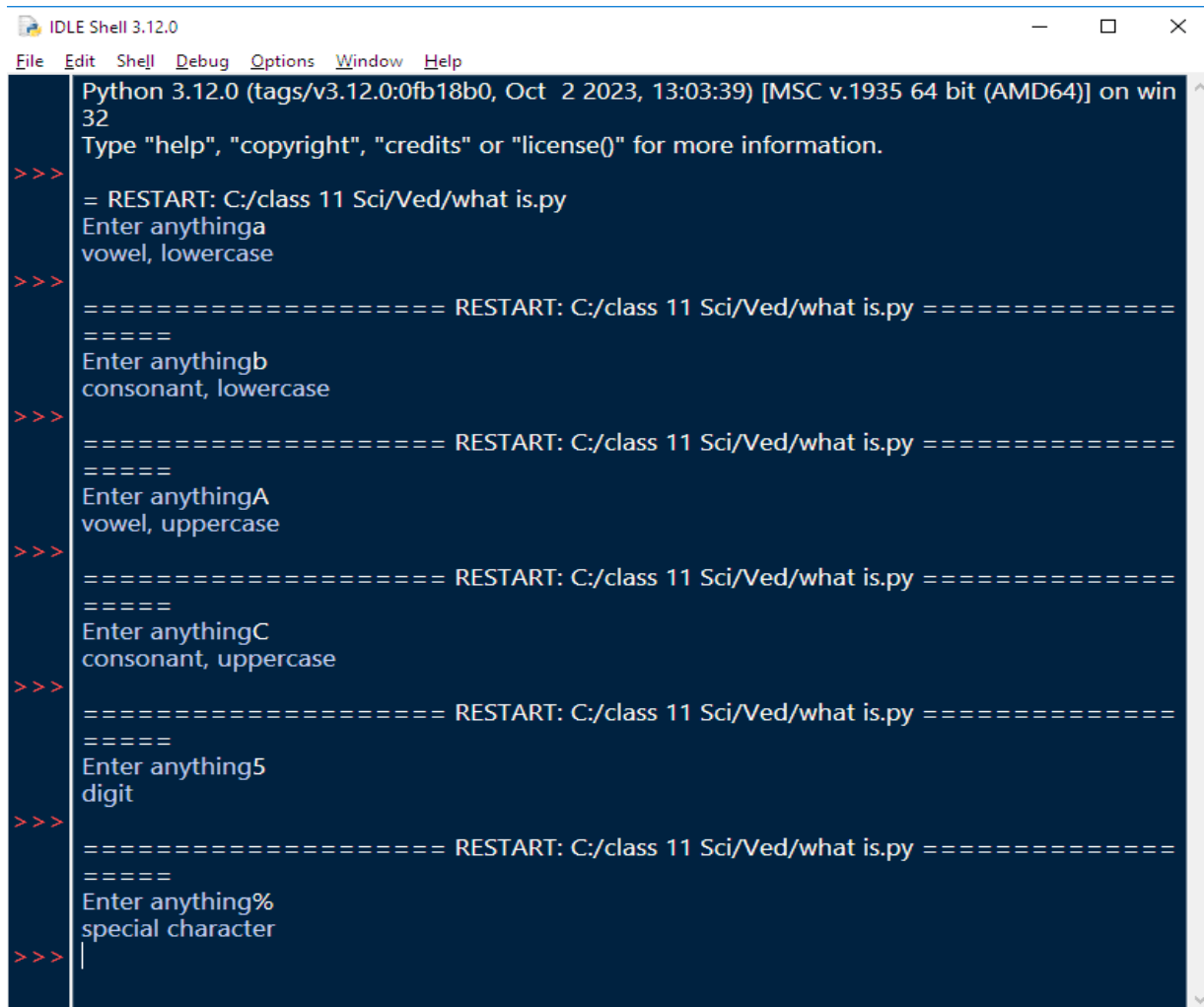
Code:-

```
a=input("Enter anything")
if a>="A" and a<="Z":
    if a=="A" or "E" or "I" or "O" or "U":
        print("vowel, uppercase")
    else:
        print("consonant, uppercase")
elif a>="a" and a<="z":
    if a=="a" or "e" or "i" or "o" or "u":
        print("vowel, lowercase")
    else:
        print("consonant, lowercase")
```

```

elif a>="0" and a<="9":
    print("digit")
else:
    print("special character")

```



```

IDLE Shell 3.12.0
File Edit Shell Debug Options Window Help
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win
32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/class 11 Sci/Ved/what is.py
Enter anything
a
vowel, lowercase
>>>
===== RESTART: C:/class 11 Sci/Ved/what is.py =====
=====
Enter anything
b
consonant, lowercase
>>>
===== RESTART: C:/class 11 Sci/Ved/what is.py =====
=====
Enter anything
A
vowel, uppercase
>>>
===== RESTART: C:/class 11 Sci/Ved/what is.py =====
=====
Enter anything
C
consonant, uppercase
>>>
===== RESTART: C:/class 11 Sci/Ved/what is.py =====
=====
Enter anything
5
digit
>>>
===== RESTART: C:/class 11 Sci/Ved/what is.py =====
=====
Enter anything
%
special character
>>>

```

Program #7-

A telephone company charges for using telephone from their consumers according to the calls made (per month) as per the given tariff:

Numbers of calls	Charge
Up to 50 calls	No charge (free)
For next 100 calls exceeding 50	call Rs. 2 per call
For next 200 calls exceeding 150	call Rs. 1.50 per call + Previous 100 calls bill amount
More than 350 calls	Rs. 1 per call + Previous 200 calls bill amount + previous 100 calls bill amount

Code:-

```

n=int(input("Enter no. of calls made"))
if n<0:
    print("You have entered wrong number of calls, please enter
correct value")
elif n<=50:
    print("Your bill is free")
elif n<=150:
    n=n-50
    n=n*2
    print("Your bill is",n)
elif n<=350:
    n=n-150
    b=n*1.5+200
    print("Your bill is",b)
elif n>350:
    n=n-350
    b=300+200*n
    print("Your bill is",b)

```

```

Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct  2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win
32
Type "help", "copyright", "credits" or "license()" for more information.
>>> = RESTART: C:\class 11 Sci\Ved\call bill.py
Enter no. of calls made40
Your bill is free
>>>
==== RESTART: C:\class 11 Sci\Ved\call bill.py =====
Enter no. of calls made102
Your bill is 104
>>>
==== RESTART: C:\class 11 Sci\Ved\call bill.py =====
Enter no. of calls made353
Your bill is 503
>>>
==== RESTART: C:\class 11 Sci\Ved\call bill.py =====
Enter no. of calls made500
Your bill is 650
>>> |

```

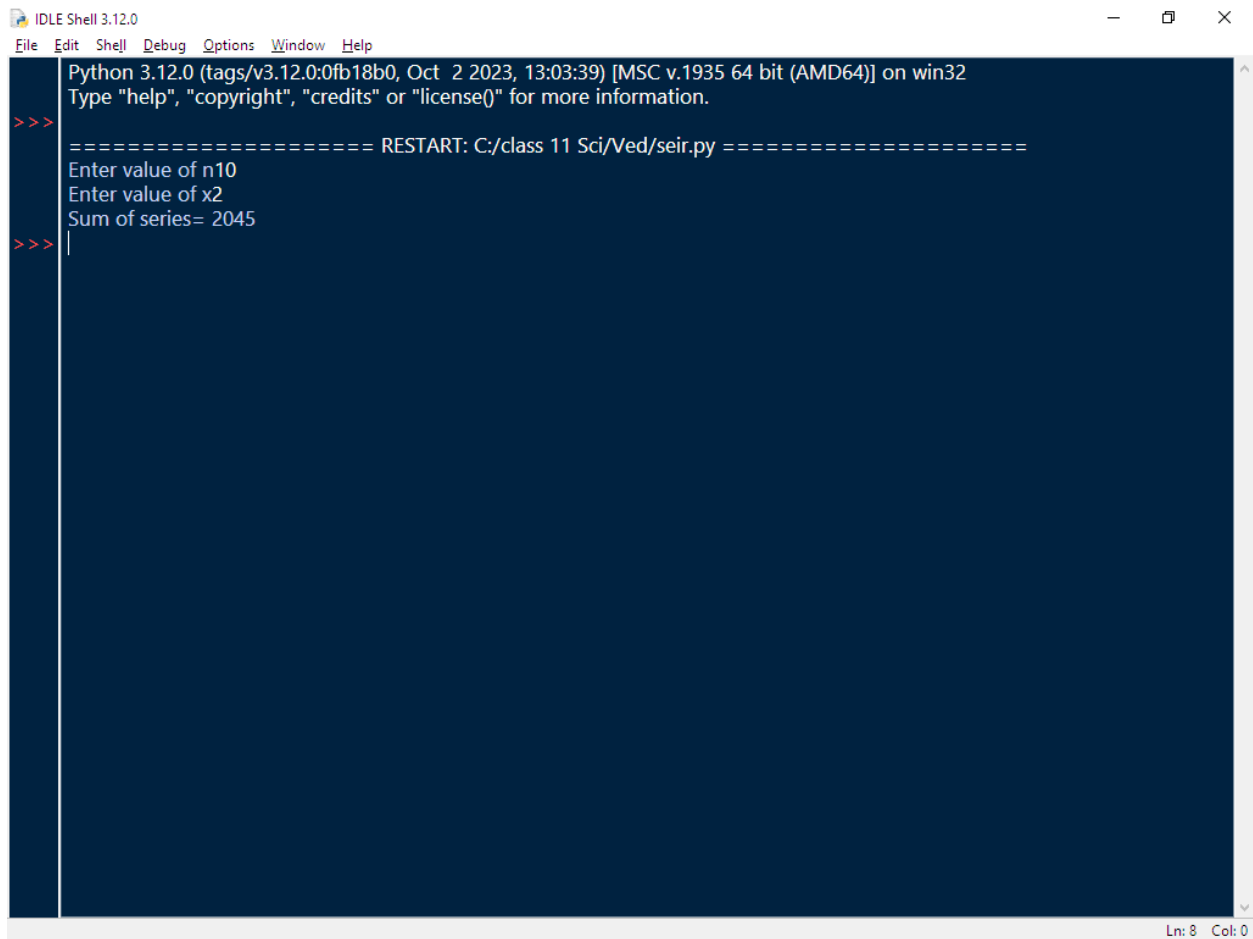
Program #8-

Write a program to input the value of x and n and print the sum of the following series:

$$\begin{aligned} &\text{➤ } 1+x+x^2+x^3+x^4+ \dots x^n \\ &\text{➤ } 1-x+x^2-x^3+x^4+ \dots x^n \\ &\text{➤ } x + \frac{x^2}{2} - \frac{x^3}{3} + \frac{x^4}{4} + \dots \frac{x^n}{n} \\ &\text{➤ } x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \frac{x^n}{n!} \end{aligned}$$

Code:-

```
n=int(input("Enter value of n"))
x=int(input("Enter value of x"))
a=1
for b in range(2,n+1):
    a=a+x**b
print("Sum of series=",a)
```



The screenshot shows a Python IDLE Shell window with the following text:

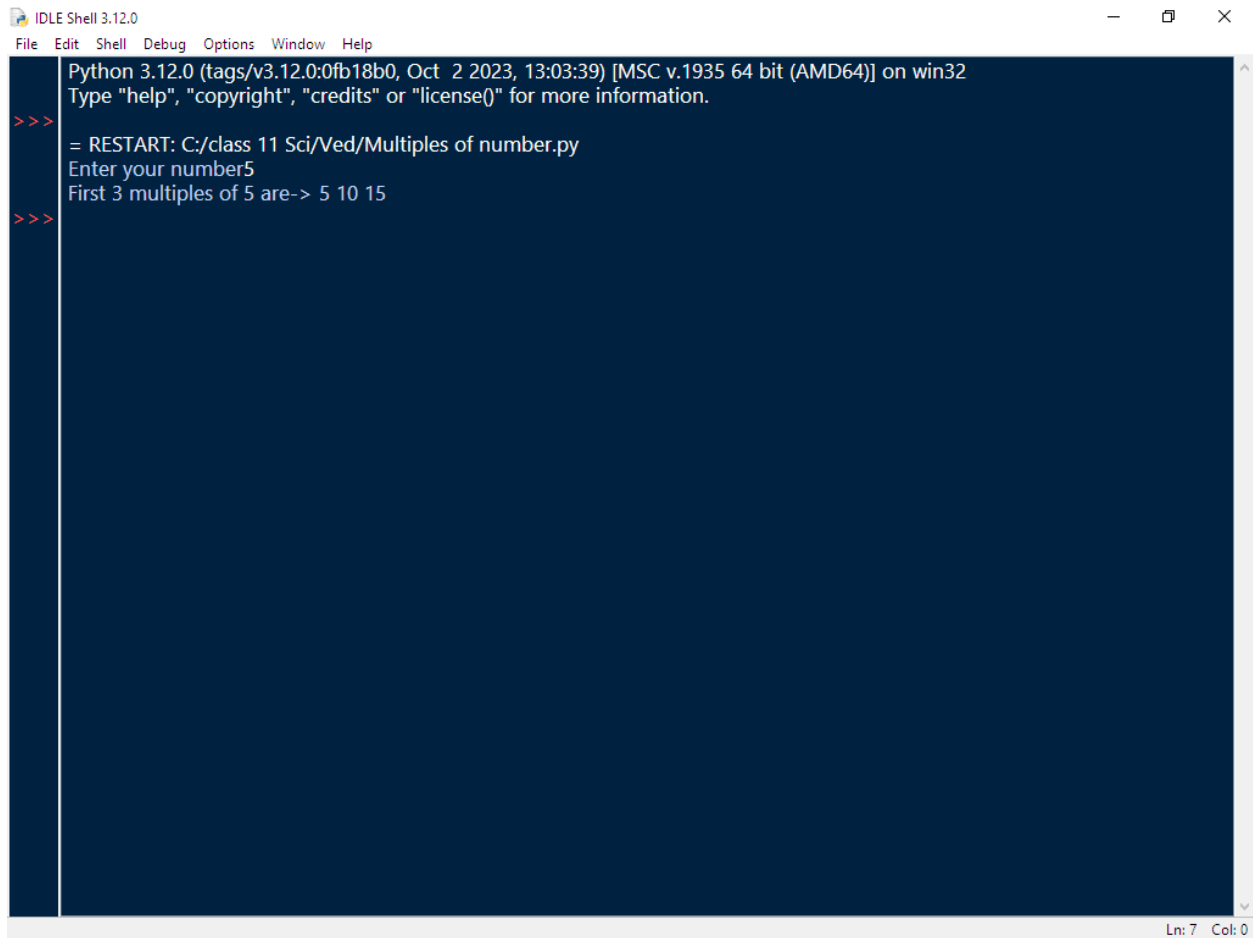
```
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/class 11 Sci/Ved/seir.py =====
Enter value of n10
Enter value of x2
Sum of series= 2045
>>>
```

Program #9-

To print first three multiples of entered number.

Code:-

```
x=int(input("Enter your number"))  
print("First 3 multiples of",x,"are->",x,x*2,x*3)
```



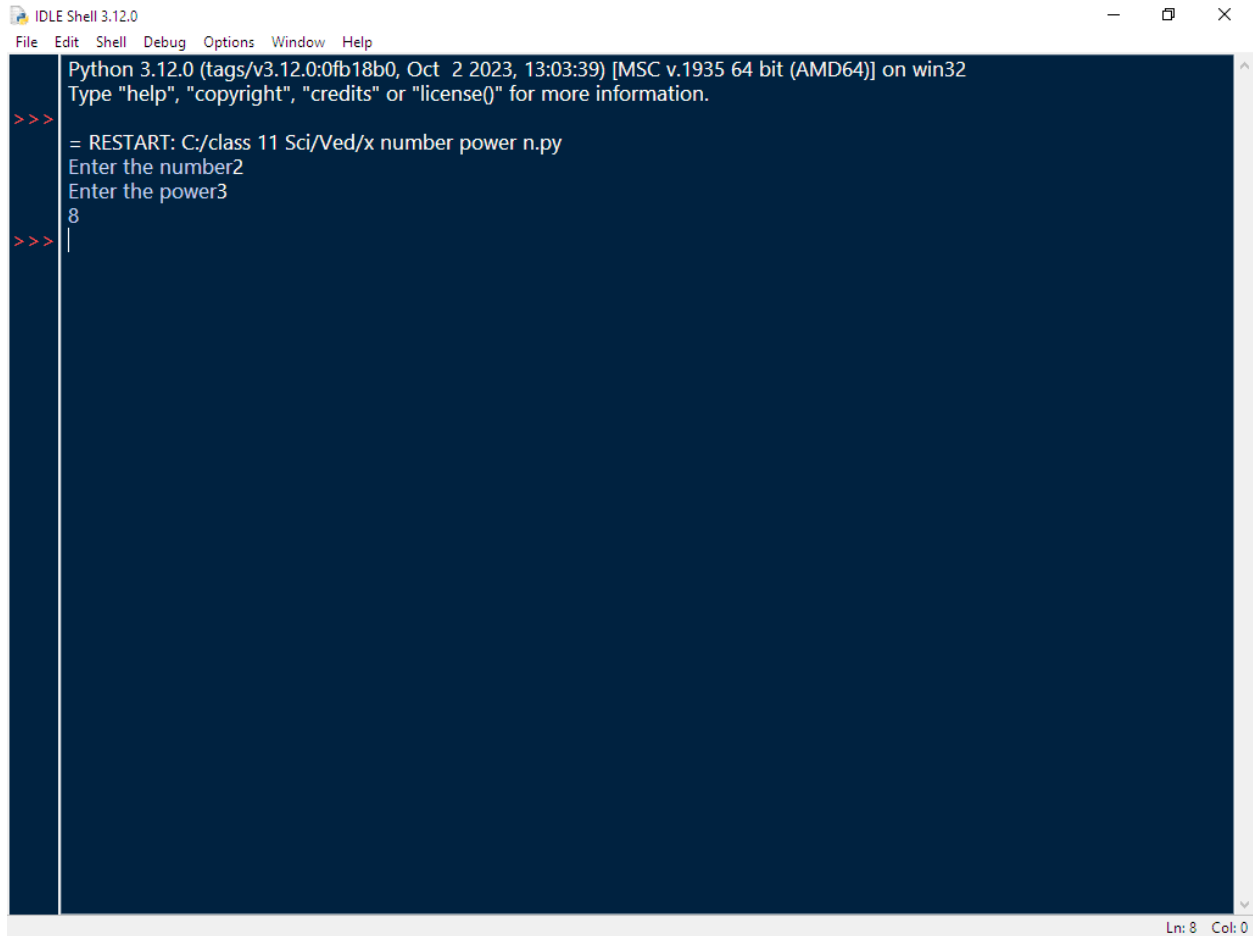
```
IDLE Shell 3.12.0  
File Edit Shell Debug Options Window Help  
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32  
Type "help", "copyright", "credits" or "license()" for more information.  
>>>  
= RESTART: C:/class 11 Sci/Ved/Multiples of number.py  
Enter your number5  
First 3 multiples of 5 are-> 5 10 15  
>>>
```

Program #10-

Input two integers x and n, compute x^n .

Code:-

```
x=int(input("Enter the number"))
n=int(input("Enter the power"))
print(x**n)
```



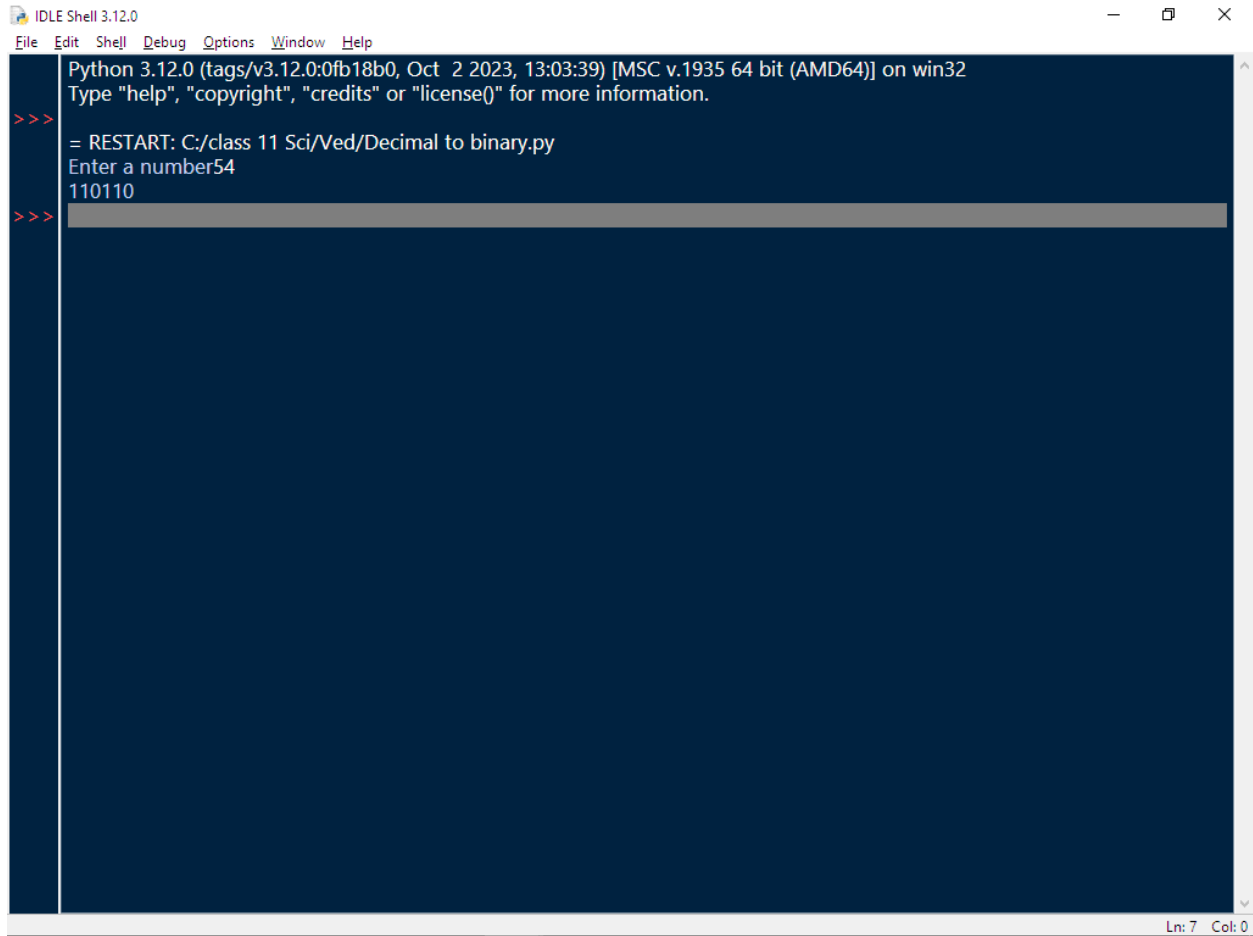
```
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/class 11 Sci/Ved/x number power n.py
Enter the number2
Enter the power3
8
>>>
```

Program #11-

Accept a decimal number and print its binary.

Code:-

```
n = int(input("Enter a number"))
b = ""
while n > 0:
    b = str(n % 2) + b
    n = n // 2
print(b)
```



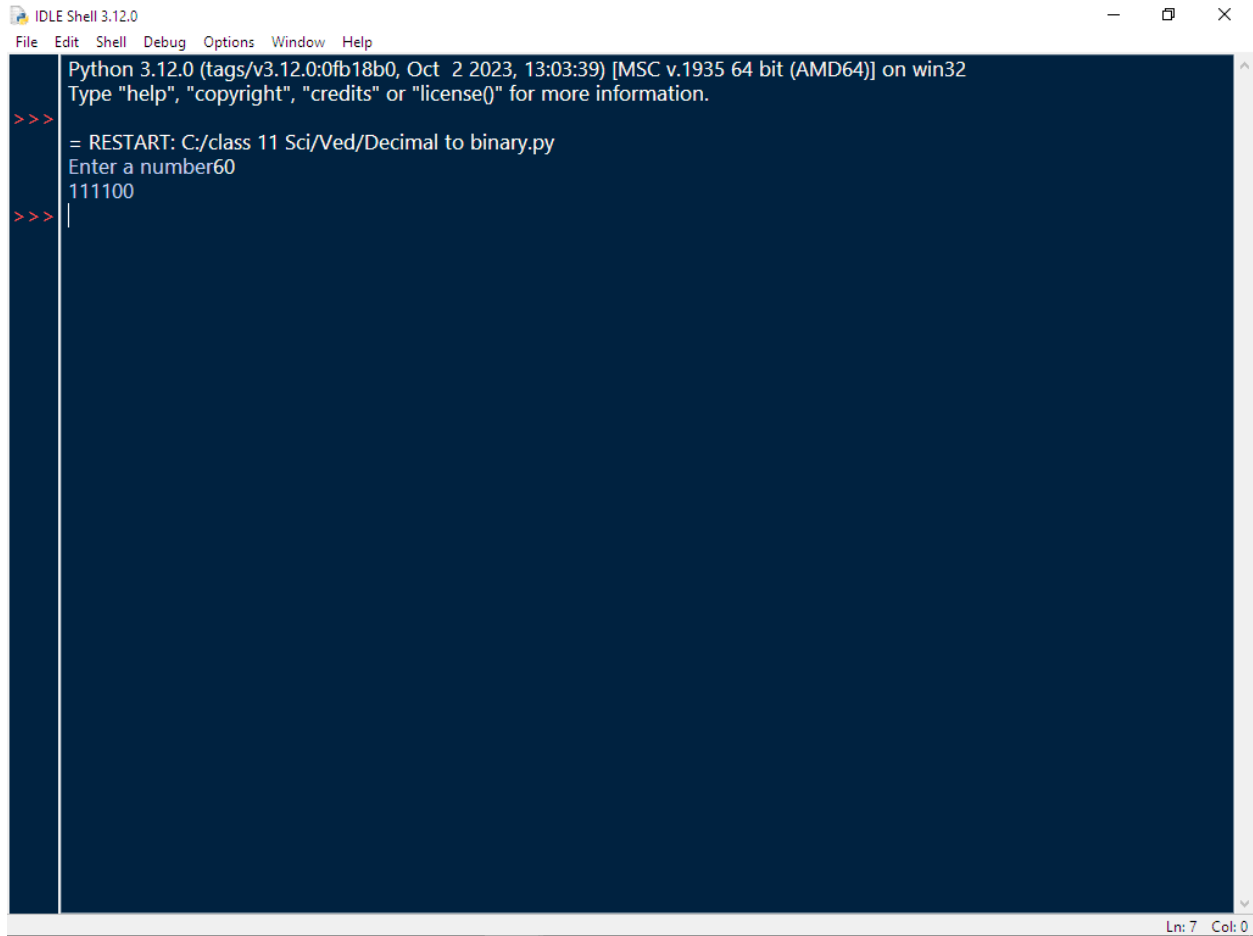
```
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/class 11 Sci/Ved/Decimal to binary.py
Enter a number54
110110
>>>
```

Program #12-

Accept a decimal number and print its octal.

Code:-

```
n = int(input("Enter a number: "))
o = ""
while n > 0:
    o = str(n % 8) + o
    n = n // 8
print(o)
```



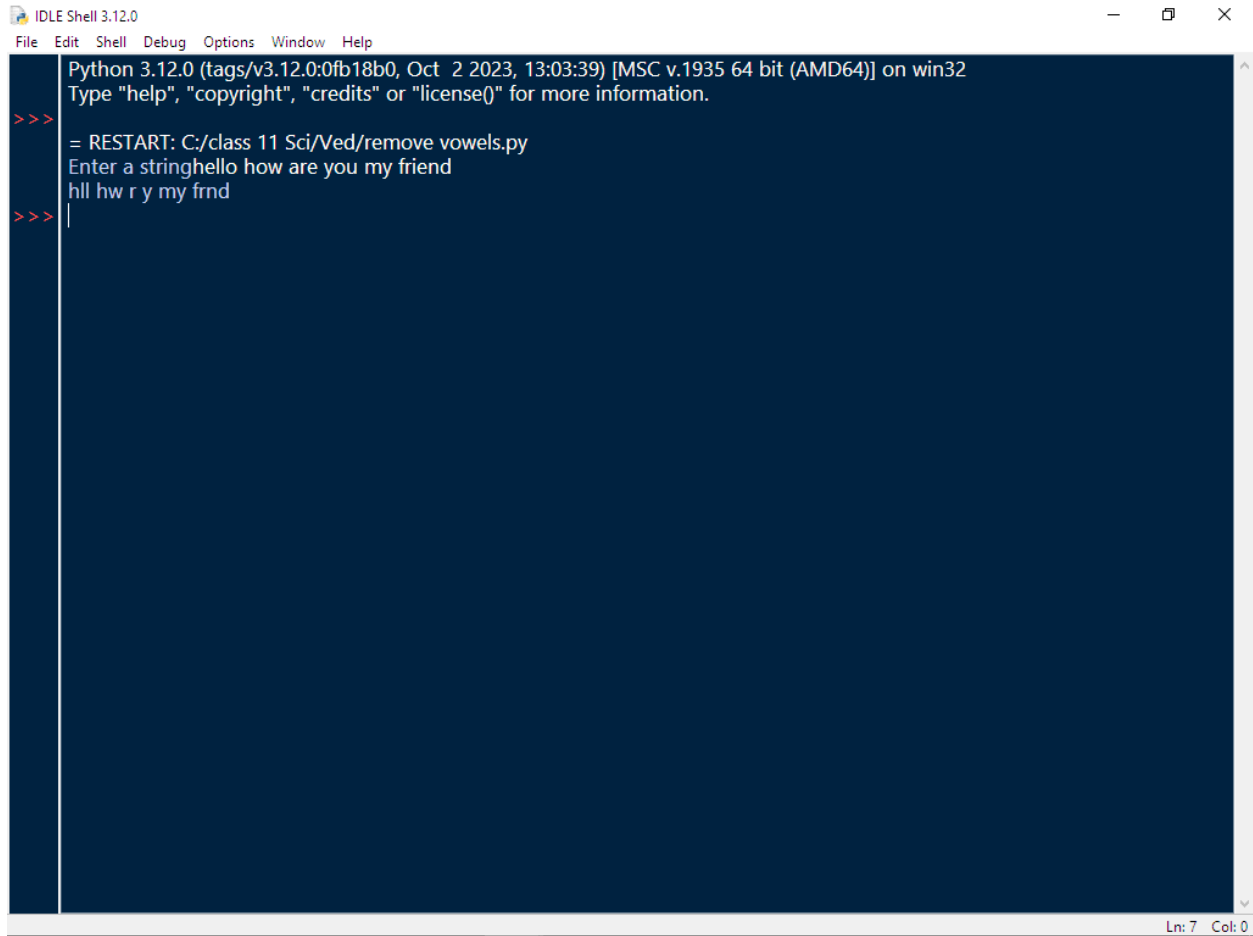
```
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/class 11 Sci/Ved/Decimal to binary.py
Enter a number60
111100
>>> |
```

Program #13-

Remove vowels from string.

Code:-

```
a=input("Enter a string")
b=""
for i in a:
    if i not in "aeiouAEIOU":
        b=b+i
print(b)
```

```
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/class 11 Sci/Ved/remove_vowels.py
Enter a stringhello how are you my friend
hll hw r y my frnd
>>> |
```

Ln: 7 Col: 0

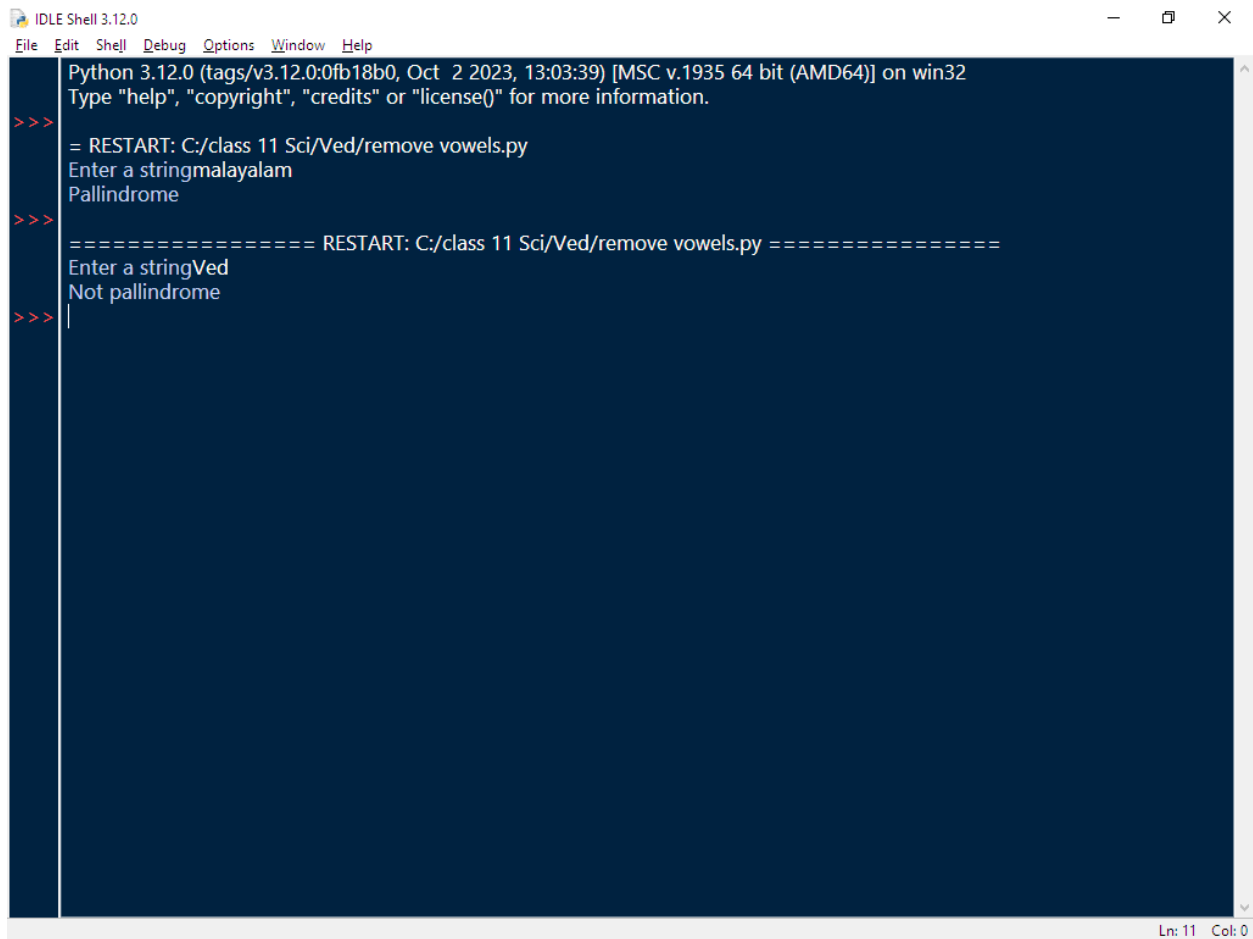
Program #14-

Check string is palindrome or not.

Code:-

```
a=input("Enter a string")
b=''
for i in range(-1,-1*len(a)-1,-1):
    b=b+a[i]
if b==a:
    print("Pallindrome")
else:
```

```
print("Not pallindrome")
```



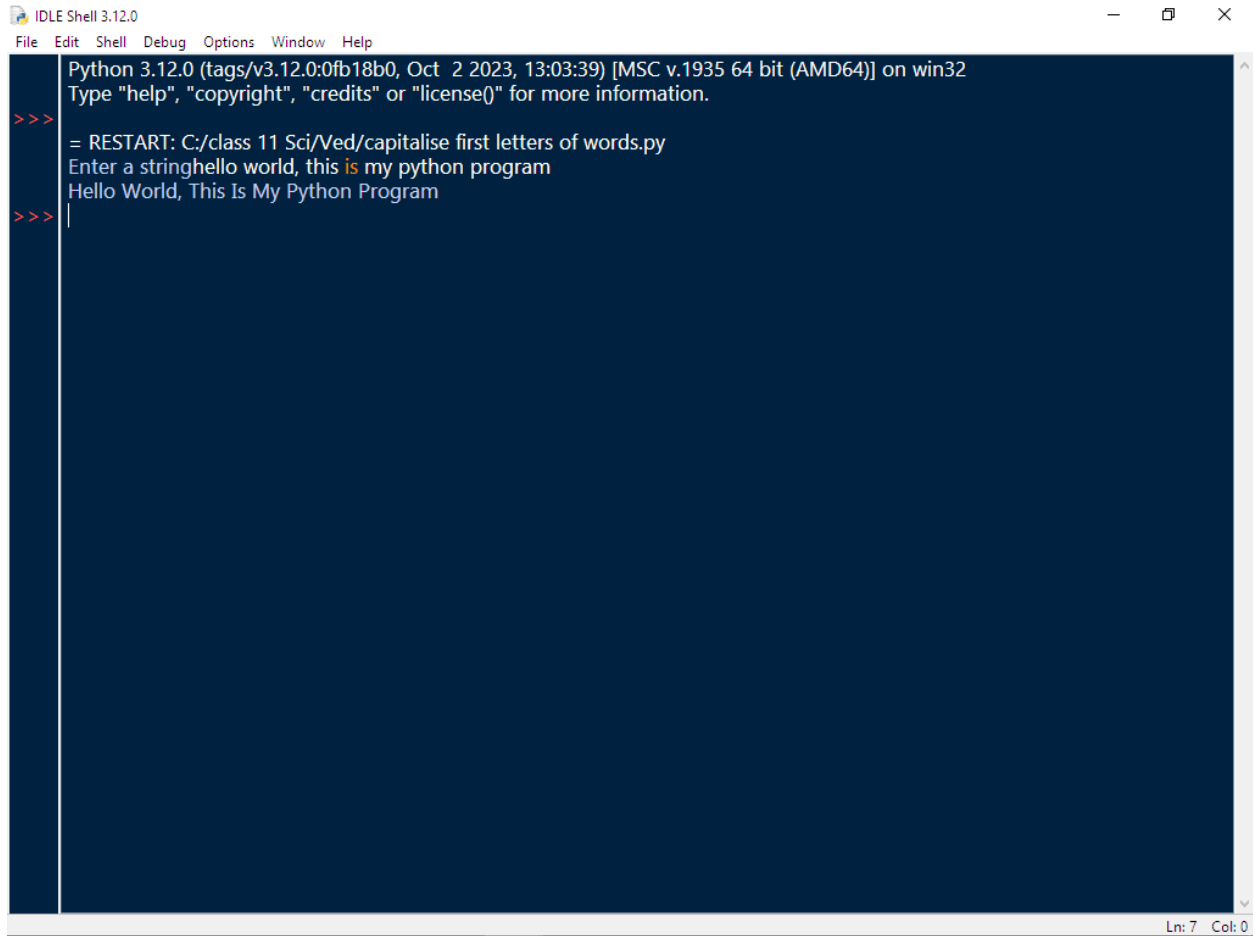
```
IDLE Shell 3.12.0
File Edit Shell Debug Options Window Help
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/class 11 Sci/Ved/remove vowels.py
Enter a stringmalayalam
Pallindrome
>>>
===== RESTART: C:/class 11 Sci/Ved/remove vowels.py =====
Enter a stringVed
Not pallindrome
>>> |
```

Program #15-

Read a line and capitalize the first letter of each word and form a new string.

Code:-

```
a=input("Enter a string")
b=a.title()
print(b)
```



The screenshot shows the IDLE Shell 3.12.0 window. The title bar reads "IDLE Shell 3.12.0". The menu bar includes "File", "Edit", "Shell", "Debug", "Options", "Window", and "Help". The main text area displays the following output:

```
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/class 11 Sci/Ved/capitalise first letters of words.py
Enter a stringhello world, this is my python program
Hello World, This Is My Python Program
>>> |
```

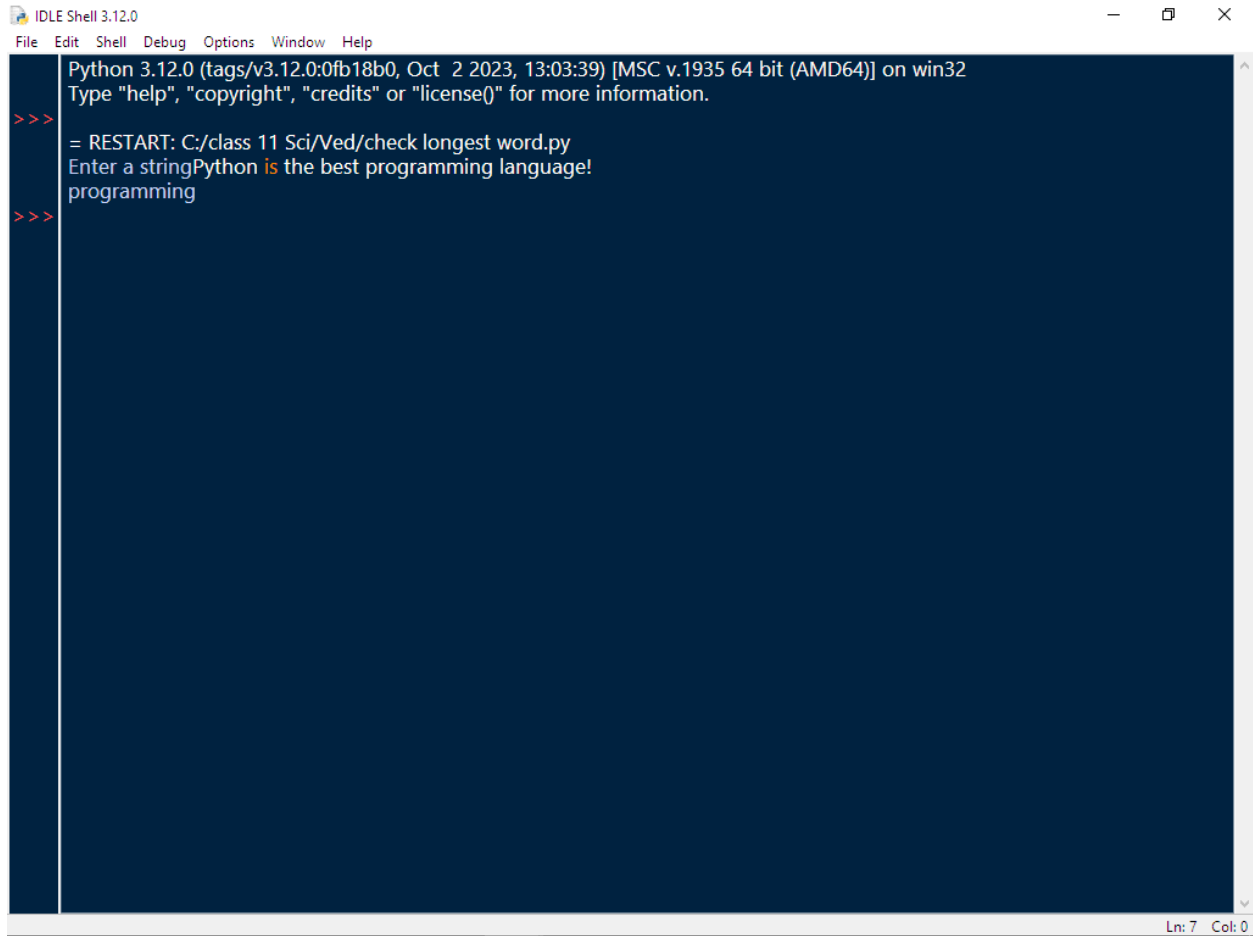
The status bar at the bottom right indicates "Ln: 7 Col: 0".

Program #16-

Read a string and display longest substring.

Code:-

```
a=input("Enter a string")
b=a.split()
c=''
for i in b:
    if len(i)>=len(c):
        c=i
print(c)
```



The screenshot shows a window titled "IDLE Shell 3.12.0" with a menu bar (File, Edit, Shell, Debug, Options, Window, Help). The shell displays the following text:

```
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/class 11 Sci/Ved/check longest word.py
Enter a stringPython is the best programming language!
programming
>>>
```

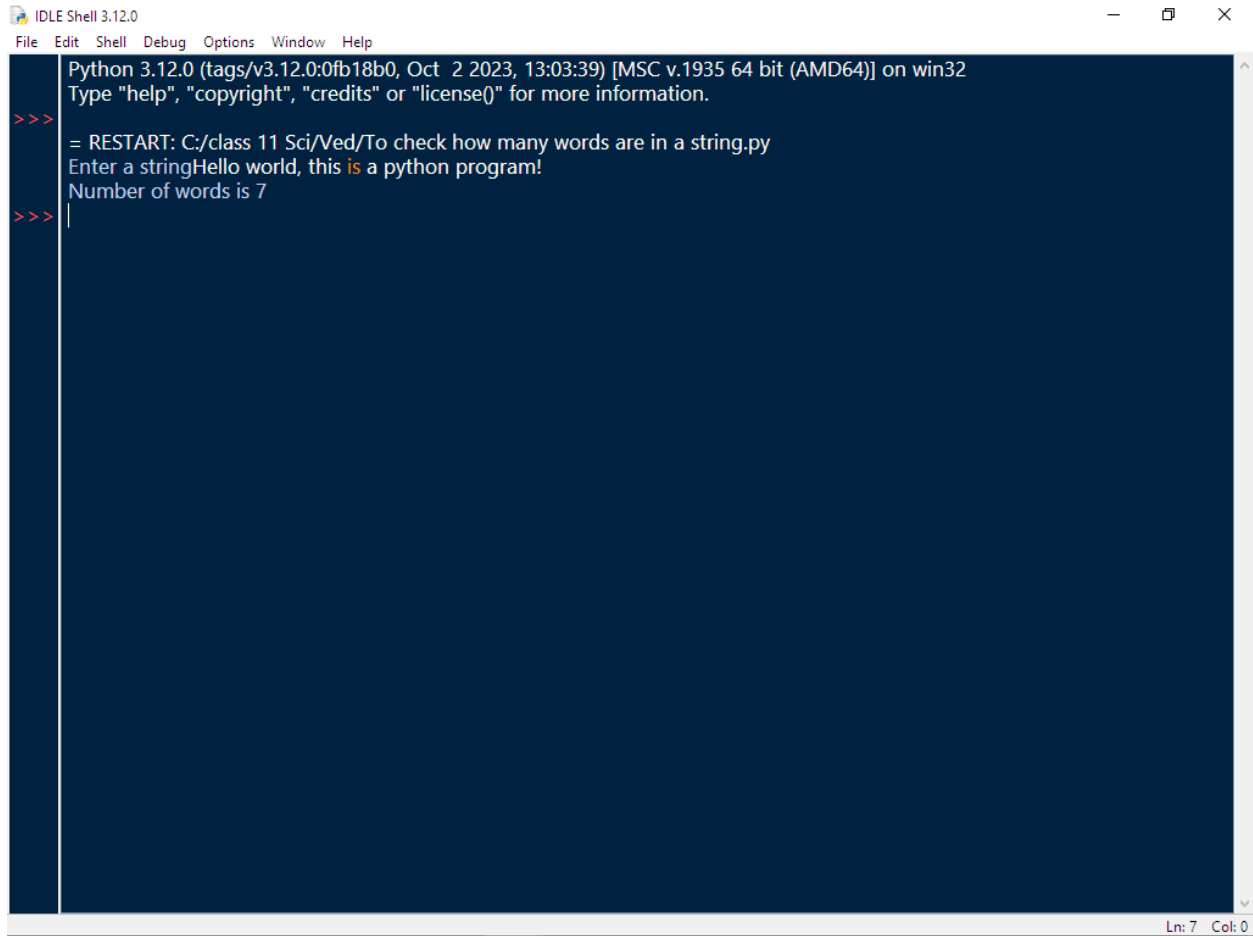
The status bar at the bottom right indicates "Ln: 7 Col: 0".

Program #17-

Count number of words in a line.

Code:-

```
a=input("Enter a string")
b=a.split()
print("Number of words is", len(b))
```



The screenshot shows the IDLE Shell 3.12.0 window. The title bar reads 'IDLE Shell 3.12.0'. The menu bar includes 'File', 'Edit', 'Shell', 'Debug', 'Options', 'Window', and 'Help'. The main text area has a dark blue background and contains the following text: 'Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32', 'Type "help", "copyright", "credits" or "license()" for more information.', a prompt '>>>', a message '= RESTART: C:/class 11 Sci/Ved/To check how many words are in a string.py', a prompt 'Enter a string:', the input 'Hello world, this is a python program!', and the output 'Number of words is 7'. Below this, another prompt '>>>' is followed by a vertical cursor. The status bar at the bottom right shows 'Ln: 7 Col: 0'.

```
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/class 11 Sci/Ved/To check how many words are in a string.py
Enter a string:Hello world, this is a python program!
Number of words is 7
>>> |
```

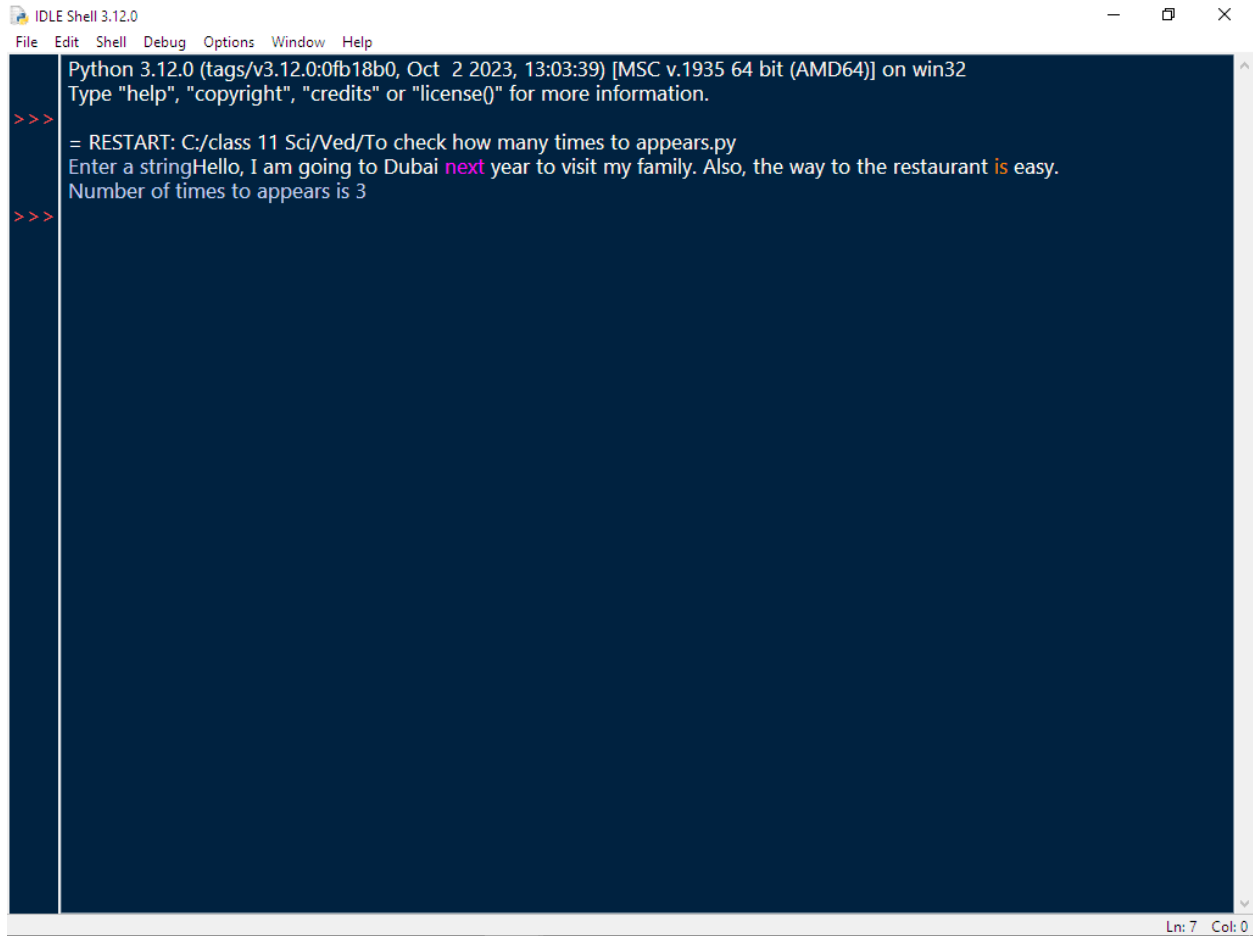
Ln: 7 Col: 0

Program #18-

Read a line and count how many time "to" appear in line.

Code:-

```
a=input("Enter a string")
b=a.split()
print("Number of times to appears is", b.count("to"))
```



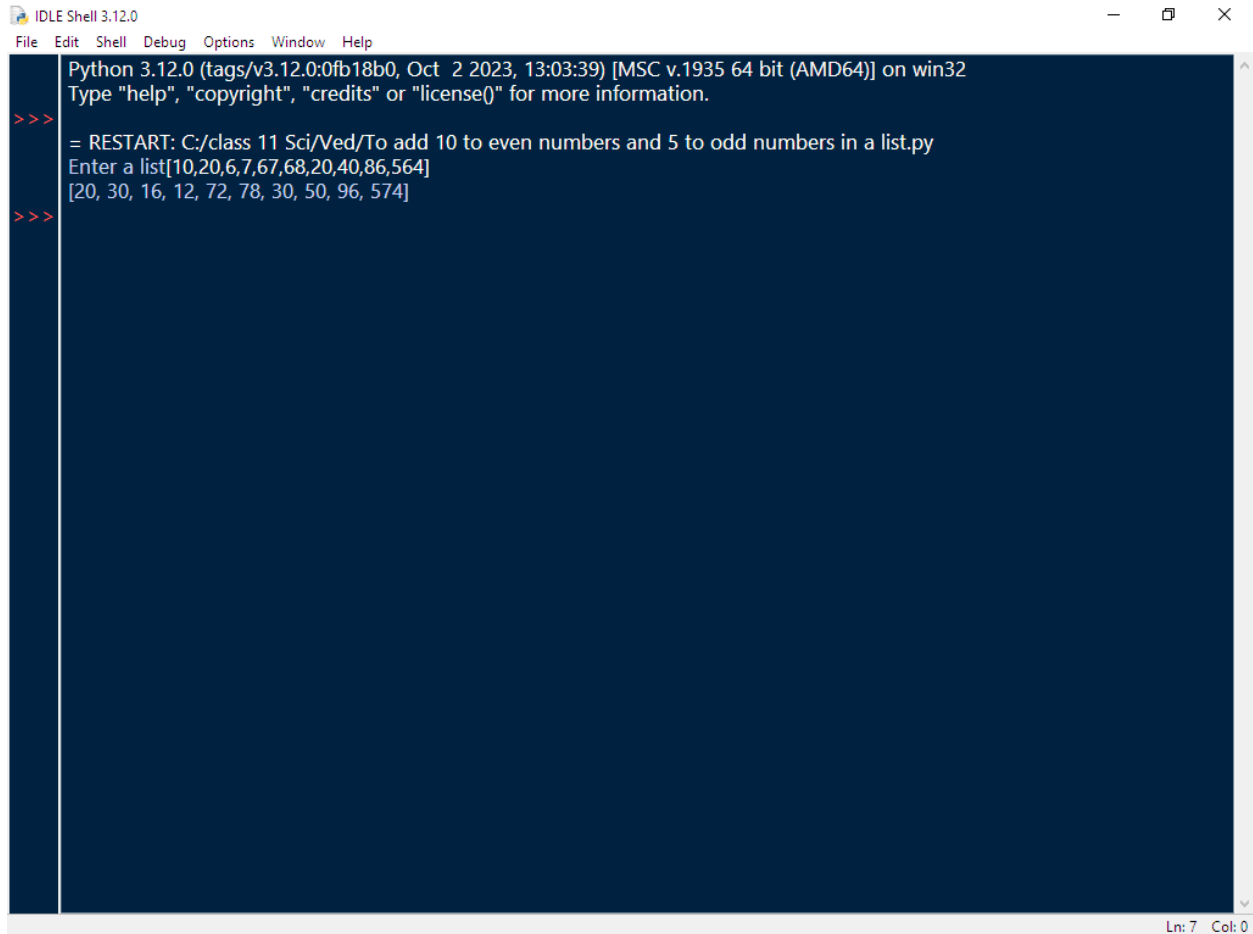
```
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/class 11 Sci/Ved/To check how many times to appears.py
Enter a stringHello, I am going to Dubai next year to visit my family. Also, the way to the restaurant is easy.
Number of times to appears is 3
>>>
```

Program #19-

Add 5 in odd values of a list and add 10 in even values of list.

Code:-

```
a=eval(input("Enter a list"))
for i in range(len(a)):
    if a[i]%2==0:
        a[i]=a[i]+10
    else:
        a[i]=a[i]+5
print(a)
```



The screenshot shows the IDLE Shell 3.12.0 window. The title bar reads 'IDLE Shell 3.12.0'. The menu bar includes 'File', 'Edit', 'Shell', 'Debug', 'Options', 'Window', and 'Help'. The main text area has a dark blue background. It displays the following text: 'Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32', 'Type "help", "copyright", "credits" or "license()" for more information.', and a prompt '>>>'. Below the prompt, it shows '= RESTART: C:/class 11 Sci/Ved/To add 10 to even numbers and 5 to odd numbers in a list.py', 'Enter a list[10,20,6,7,67,68,20,40,86,564]', and the resulting list '[20, 30, 16, 12, 72, 78, 30, 50, 96, 574]'. A second prompt '>>>' is visible. The status bar at the bottom right shows 'Ln: 7 Col: 0'.

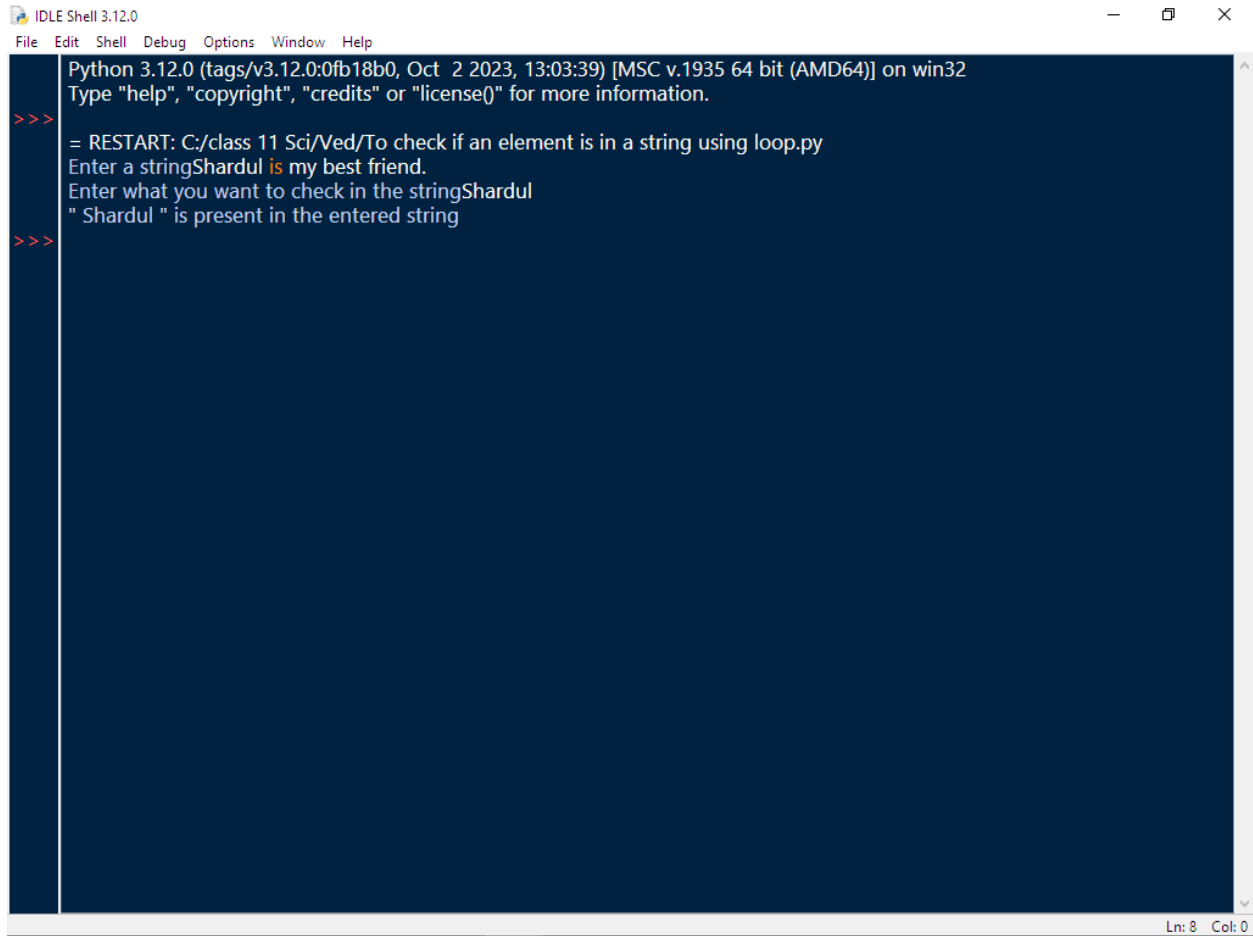
```
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/class 11 Sci/Ved/To add 10 to even numbers and 5 to odd numbers in a list.py
Enter a list[10,20,6,7,67,68,20,40,86,564]
[20, 30, 16, 12, 72, 78, 30, 50, 96, 574]
>>>
```

Program #20-

Write a program to demonstrate Linear Search(search an element present in list using loop)

Code:-

```
a=input("Enter a string")
b=a.split()
c=input("Enter what you want to check in the string")
for i in b:
    if c==i:
        print('','',c,'',"is present in the entered string")
        break
```



```
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/class 11 Sci/Ved/To check if an element is in a string using loop.py
Enter a stringShardul is my best friend.
Enter what you want to check in the stringShardul
" Shardul " is present in the entered string
>>>
```

Program #21-

Display frequencies of all elements of a list.

Code:-

```
a=eval(input("Enter a list"))
for i in a:
    print(i,"occurs",a.count(i),"times")
```



```
IDLE Shell 3.12.0
File Edit Shell Debug Options Window Help
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/class 11 Sci/Ved/Count frequency of elements in a list.py
Enter a list["Hello","How","Are","how",2,4,56,6,3,7,3,1,3,5,7,32,4,7,3]
Hello occurs 1 times
How occurs 1 times
Are occurs 1 times
how occurs 1 times
2 occurs 1 times
4 occurs 2 times
56 occurs 1 times
6 occurs 1 times
3 occurs 4 times
7 occurs 3 times
3 occurs 4 times
1 occurs 1 times
3 occurs 4 times
5 occurs 1 times
7 occurs 3 times
32 occurs 1 times
4 occurs 2 times
7 occurs 3 times
3 occurs 4 times
>>> |
```

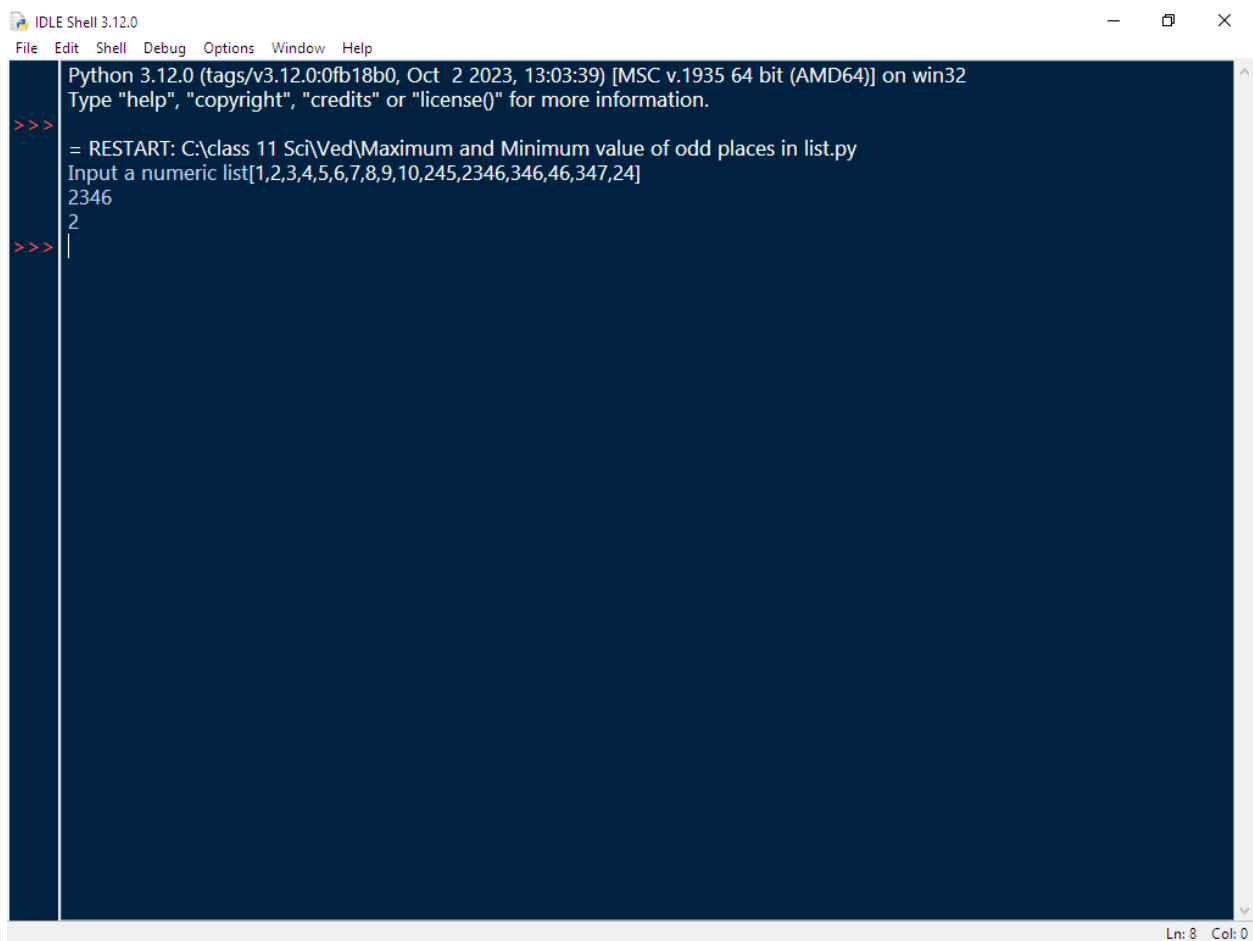
Program #22-

Find the largest/smallest number in a list elements at the **odd** location.

Code:-

```
a=eval(input("Input a numeric list"))
b=[]
for i in range(1,len(a),2):
    b.append(a[i])
c=0
d=0
```

```
for j in b:
    if j>=c:
        c=j
print(c)
for k in b:
    if k<=c:
        c=k
print(c)
```



The screenshot shows the IDLE Shell 3.12.0 window. The title bar reads "IDLE Shell 3.12.0". The menu bar includes "File", "Edit", "Shell", "Debug", "Options", "Window", and "Help". The shell area has a dark blue background with white text. It displays the Python 3.12.0 startup message: "Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32". Below this, it says "Type 'help', 'copyright', 'credits' or 'license()' for more information." The prompt ">>>" is followed by the text "= RESTART: C:\class 11 Sci\Ved\Maximum and Minimum value of odd places in list.py". Then, it says "Input a numeric list[1,2,3,4,5,6,7,8,9,10,245,2346,346,46,347,24]". The user has entered "2346" and "2" on separate lines. The prompt ">>>" is followed by a vertical bar "|". The status bar at the bottom right shows "Ln: 8 Col: 0".

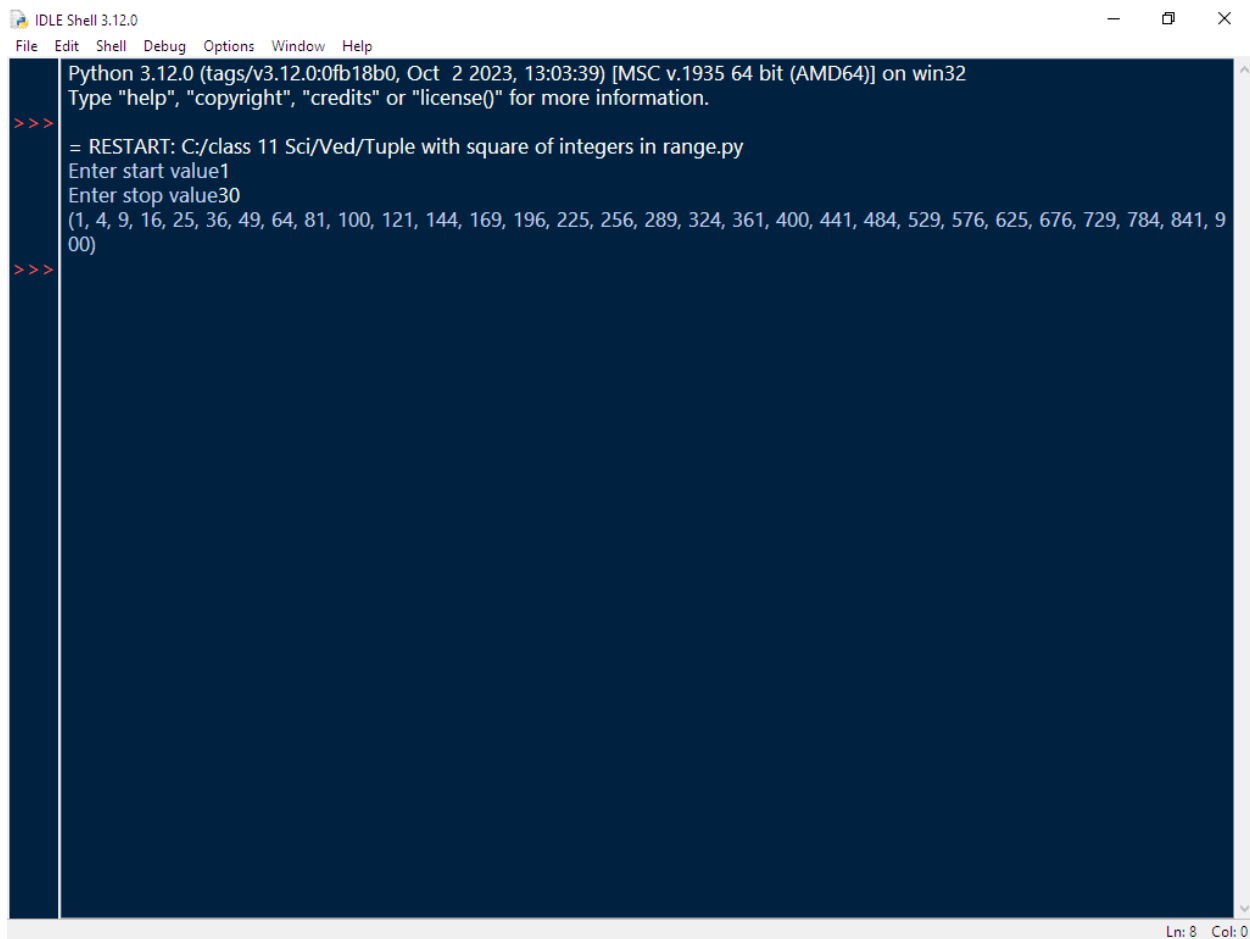
```
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\class 11 Sci\Ved\Maximum and Minimum value of odd places in list.py
Input a numeric list[1,2,3,4,5,6,7,8,9,10,245,2346,346,46,347,24]
2346
2
>>> |
```

Program #23-

Create a tuple containing the squares of the integers in given range.

Code:-

```
a=()
b=int(input("Enter start value"))
c=int(input("Enter stop value"))
for i in range(b,c+1):
    a+=(i**2,)
print(a)
```



The screenshot shows the IDLE Shell 3.12.0 interface. The title bar reads "IDLE Shell 3.12.0". The menu bar includes "File", "Edit", "Shell", "Debug", "Options", "Window", and "Help". The shell area displays the following text:

```
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/class 11 Sci/Ved/Tuple with square of integers in range.py
Enter start value1
Enter stop value30
(1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, 256, 289, 324, 361, 400, 441, 484, 529, 576, 625, 676, 729, 784, 841, 900)
>>>
```

The status bar at the bottom right indicates "Ln: 8 Col: 0".

Program #24-

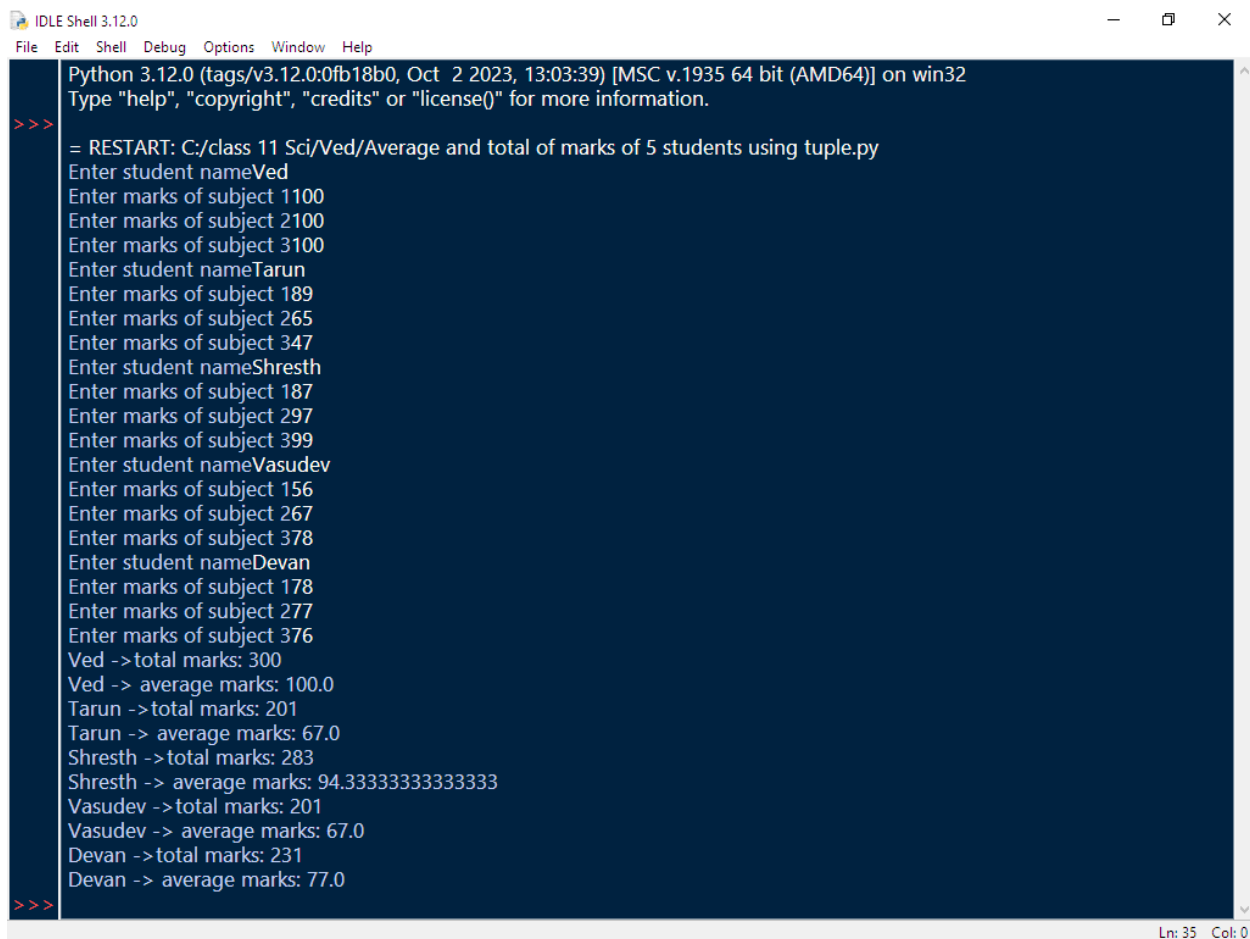
Create a nested tuple to store marks of three subject for 5 students
print total marks and average of each student.

Code:-

```

f=()
for i in range(5):
    a=input("Enter student name")
    b=int(input("Enter marks of subject 1"))
    c=int(input("Enter marks of subject 2"))
    d=int(input("Enter marks of subject 3"))
    e=(a,b,c,d)
    f=f+(e,)
total=0
for j in f:
    total=j[1]+j[2]+j[3]
    print(j[0],"->total marks:",total)
    print(j[0],"-> average marks:",total/3)
    total=0

```



```

Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/class 11 Sci/Ved/Average and total of marks of 5 students using tuple.py
Enter student nameVed
Enter marks of subject 1100
Enter marks of subject 2100
Enter marks of subject 3100
Enter student nameTarun
Enter marks of subject 189
Enter marks of subject 265
Enter marks of subject 347
Enter student nameShresth
Enter marks of subject 187
Enter marks of subject 297
Enter marks of subject 399
Enter student nameVasudev
Enter marks of subject 156
Enter marks of subject 267
Enter marks of subject 378
Enter student nameDevan
Enter marks of subject 178
Enter marks of subject 277
Enter marks of subject 376
Ved ->total marks: 300
Ved -> average marks: 100.0
Tarun ->total marks: 201
Tarun -> average marks: 67.0
Shresth ->total marks: 283
Shresth -> average marks: 94.33333333333333
Vasudev ->total marks: 201
Vasudev -> average marks: 67.0
Devan ->total marks: 231
Devan -> average marks: 77.0
>>>

```

Program #25-

Area and perimeter of circle/triangle/square/rectangle.(give choice)

Code:-

```

n=1
z="yes"

```

```

while n>0:
    if z.lower()=="yes":
        a=input("Enter a shape you want to find area and perimeter of-
>(circle/triangle/square/rectangle)")
        if a.lower()=="circle":
            b=int(input("Enter the radius of circle"))
            print("Perimeter is->",b*2*22/7)
            print("Area is->",b*b*22/7)
        elif a.lower()=="triangle":
            b=int(input("Enter first side of triangle"))
            c=int(input("Enter second side of triangle"))
            d=int(input("Enter third side of triangle"))
            s=b+c+d/2
            area=(s*(s-b)*(s-c)*(s-d))**(1/2)
            print("Perimeter is->",b+c+d)
            print("Area is->",area)
        elif a.lower()=="square":
            b=int(input("Enter side of square"))
            print("Perimeter is->",4*b)
            print("Area is->",b**2)
        elif a.lower()=="rectangle":
            b=int(input("Enter length"))
            c=int(input("Enter breadth"))
            print("Perimeter is->",2*(b+c))
            print("Area is->",b*c)
        else:
            print("Wrong shape")
            z=input("Would you like to continue? (yes/no)")
    else:
        n=0

```

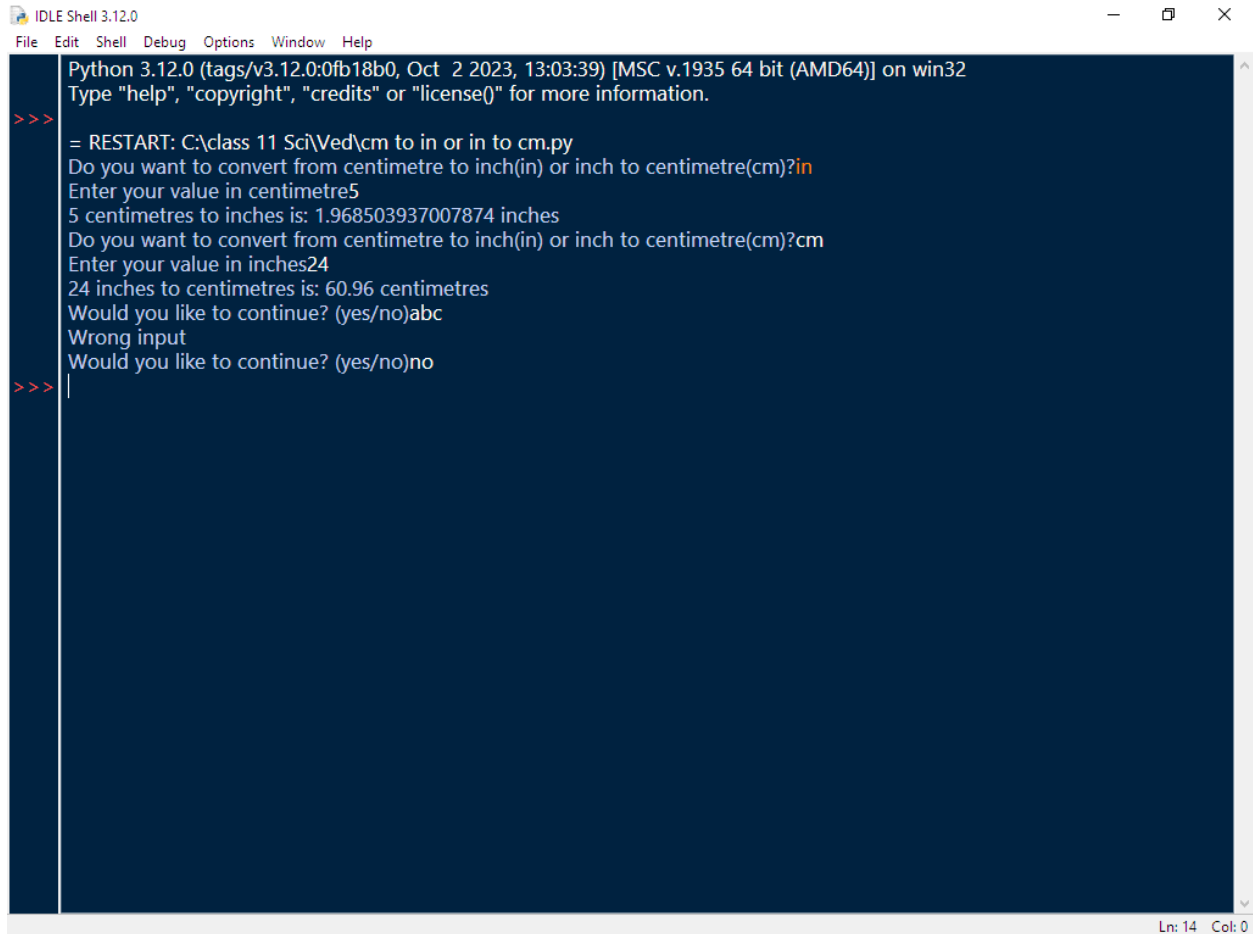
```
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
==== RESTART: C:\class 11 Sci\Ved\Choose shape to find area and perimeter.py ====
Enter a shape you want to find area and perimeter of->(circle/triangle/square/rectangle)CirCle
Enter the radius of circle5
Perimeter is-> 31.428571428571427
Area is-> 78.57142857142857
Would you like to continue? (yes/no)yes
Enter a shape you want to find area and perimeter of->(circle/triangle/square/rectangle)rectangle
Enter length36
Enter breadth67
Perimeter is-> 206
Area is-> 2412
Would you like to continue? (yes/no)no
>>>
```

Program #26-

Convert Length Centimeter to inches and inches to centimeter.(give choice)

Code:-

```
a=input("Do you want to convert from centimetre to inch(in) or inch to centimetre(cm)?")
if a.lower()=="in":
    b=int(input("Enter your value in centimetre"))
    print(b,"centimetres to inches is:",b/2.54,"inches")
elif a.lower()=="cm":
    b=int(input("Enter your value in inches"))
    print(b,"inches to centimetres is:",b*2.54,"centimetres")
```



The screenshot shows the IDLE Shell 3.12.0 interface. The title bar reads 'IDLE Shell 3.12.0'. The menu bar includes 'File', 'Edit', 'Shell', 'Debug', 'Options', 'Window', and 'Help'. The shell area has a dark blue background with white text. It shows the following interaction:

```
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\class 11 Sci\Ved\cm to in or in to cm.py
Do you want to convert from centimetre to inch(in) or inch to centimetre(cm)?in
Enter your value in centimetre5
5 centimetres to inches is: 1.968503937007874 inches
Do you want to convert from centimetre to inch(in) or inch to centimetre(cm)?cm
Enter your value in inches24
24 inches to centimetres is: 60.96 centimetres
Would you like to continue? (yes/no)abc
Wrong input
Would you like to continue? (yes/no)no
>>> |
```

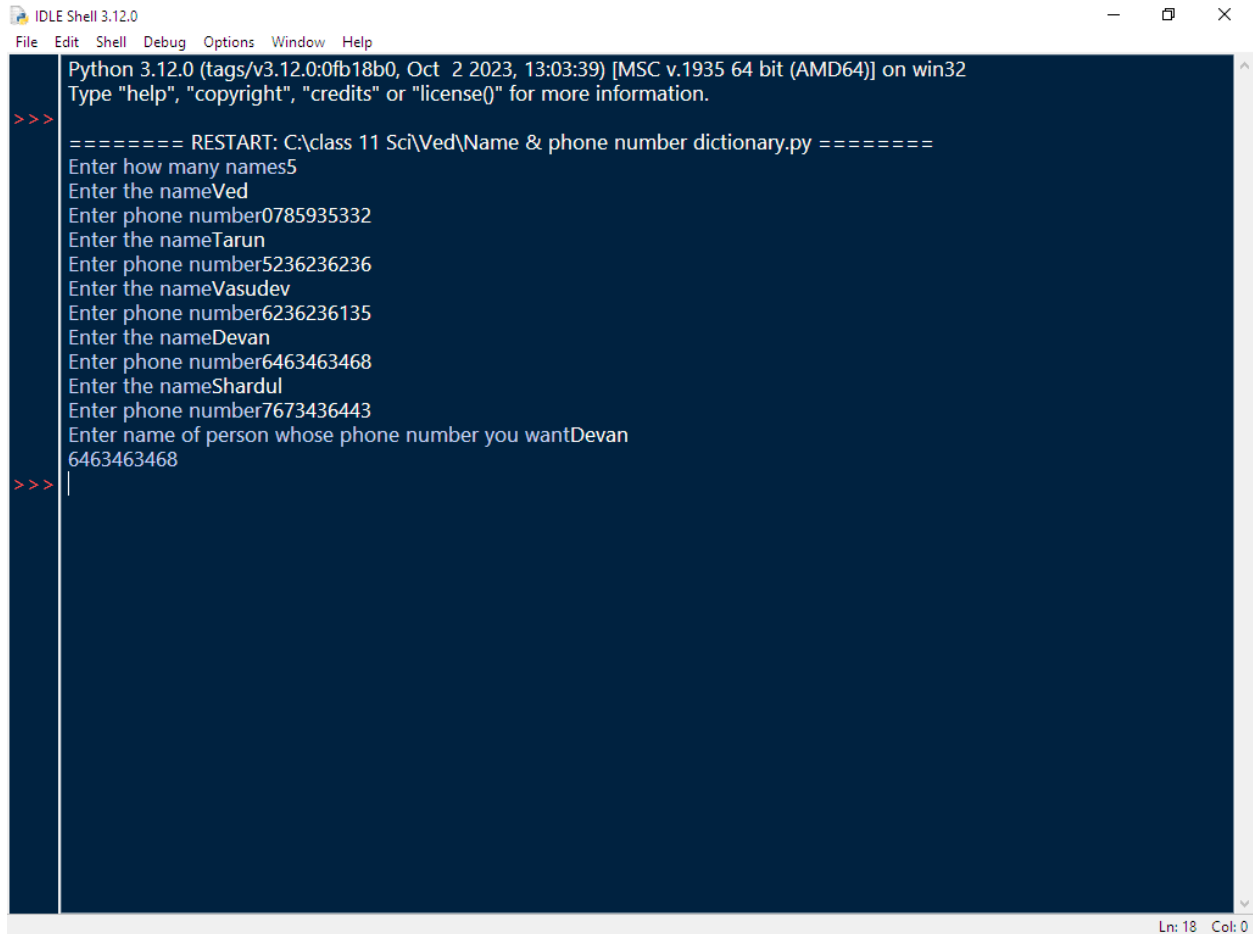
The status bar at the bottom right indicates 'Ln: 14 Col: 0'.

Program #27-

Create a dictionary, to input n names and phone numbers to store it in a dictionary and print the phone number of a particular name.

Code:-

```
d={}
n=int(input("Enter how many names"))
for i in range(n):
    a=input("Enter the name")
    b=int(input("Enter phone number"))
    d1={a:b}
    d.update(d1)
m=input("Enter name of person whose phone number you want")
print(d[m])
```



```
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:\class 11 Sci\Ved\Name & phone number dictionary.py =====
Enter how many names5
Enter the nameVed
Enter phone number0785935332
Enter the nameTarun
Enter phone number5236236236
Enter the nameVasudev
Enter phone number6236236135
Enter the nameDevan
Enter phone number6463463468
Enter the nameShardul
Enter phone number7673436443
Enter name of person whose phone number you wantDevan
6463463468
>>>
```

Program #28-

Create a dictionary, to input n names and phone numbers to store it in a dictionary and print the phone number of a particular name.

Code:-

```
d={}
n=int(input("Enter how many classes"))
for i in range(n):
    a=input("Enter the class")
    b=input("Enter name of class teacher")
    d1={a:b}
    d.update(d1)
print("Your dictionary is->", d)
m=input("Enter class whose class teacher you want")
print(d[m])
```



```
Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

>>> = RESTART: C:/class 11 Sci/Ved/Class and class teacher dictionary.py
Enter how many classes3
Enter the class11 Commerce
Enter name of class teacherShobit sir
Enter the class11 Science
Enter name of class teacherAlok sir
Enter the class1 A
Enter name of class teacherVasudev
Your dictionary is-> {'11 Commerce': 'Shobit sir', '11 Science ': 'Alok sir', '1 A': 'Vasudev'}
Enter class whose class teacher you want1 A
Vasudev
>>> |
```

Ln: 15 Col: 0